Configure IPv6 Addressing:-

Tasks:-

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Ans.:-

Windows (Command Prompt – Run as Administrator) :-

Step 1: Find the network adapter name :-

netsh interface ipv6 show interfaces

```
Idx     Met         MTU          State                Name
---  ----------  ----------  ------------  ---------------------------
  1          75  4294967295  connected     Loopback Pseudo-Interface 1
 22          50        1400  connected     Wi-Fi
 17           5        1500  disconnected  Ethernet
 18          25        1500  disconnected  Local Area Connection* 1
  8          25        1500  disconnected  Local Area Connection* 10
 39          15        1500  connected     vEthernet (Default Switch)
 14          15        1500  connected     vEthernet (internal)
  7          15        1500  disconnected  vEthernet (extrenal)
```

Step 2: Assign a static IPv6 address :-

netsh interface ipv6 add address "Wi-Fi" 2402:a00:142:7323::10/64

Step 3: Verify the configuration :-

Ipconfig

```
Ethernet adapter vEthernet (Default Switch):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::fdca:7189:f649:4306%39
   IPv4 Address. . . . . . . . . . . : 172.25.112.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (internal):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6c67:7449:3ba3:5ce1%14
   Autoconfiguration IPv4 Address. . : 169.254.62.70
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter vEthernet (extrenal):

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:a00:142:7323:cf4b:53ac:6dfd:1c78
   Temporary IPv6 Address. . . . . . : 2402:a00:142:7323:8d80:8ed0:ebf2:9f92
   Link-local IPv6 Address . . . . . : fe80::b1c8:55cf:38f0:d3b7%22
   IPv4 Address. . . . . . . . . . . : 192.168.1.7
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%22
                                       192.168.1.1
```

**2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.**

Ans. :-

Windows :-

Step 1: Configure a ULA on your Wi-Fi adapter :-

netsh interface ipv6 add address "Wi-Fi" fd00:1234:5678:1::10/64

Step 2: Verify the address :-

Ipconfig **or**

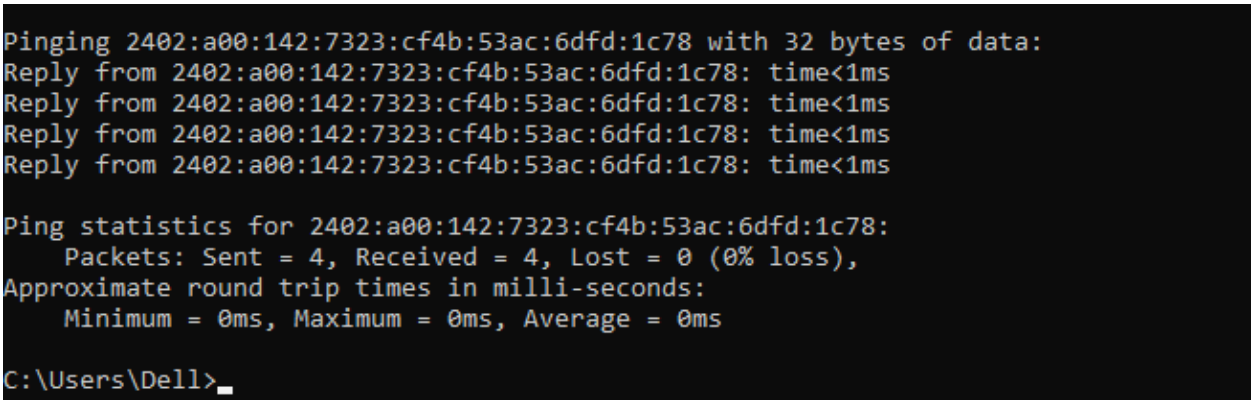
netsh interface ipv6 show address "Wi-Fi"

```
Address 2402:a00:142:7323:8d80:8ed0:ebf2:9f92 Parameters
------------------------------------------------------------
Interface Luid     : Wi-Fi
Scope Id           : 0.0
Valid Lifetime     : 23h48m25s
Preferred Lifetime : 48m25s
DAD State          : Preferred
Address Type       : Temporary
Skip as Source     : false

Address 2402:a00:142:7323:cf4b:53ac:6dfd:1c78 Parameters
------------------------------------------------------------
Interface Luid     : Wi-Fi
Scope Id           : 0.0
Valid Lifetime     : 23h48m25s
Preferred Lifetime : 48m25s
DAD State          : Preferred
Address Type       : Public
Skip as Source     : false

Address fe80::b1c8:55cf:38f0:d3b7%22 Parameters
------------------------------------------------------------
Interface Luid     : Wi-Fi
Scope Id           : 0.22
Valid Lifetime     : infinite
Preferred Lifetime : infinite
DAD State          : Preferred
Address Type       : Other
Skip as Source     : false
```

Step 3: Ping another device on your LAN :-



3. Set up a temporary IPv6 address on your system and check if it changes after a rebootornetworkdisconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

Ans.:-

IPv6 Temporary Address Test

Objective:

Configure and verify a temporary IPv6 address, then check whether it changes after a reboot or network reconnect.

Steps Performed:

1. Opened Command Prompt.

2. Ran the following command to view IPv6 addresses:

   ipconfig /all

3. Located the Temporary IPv6 Address under the active network adapter.

4. Noted the temporary IPv6 address.

5. Restarted the computer (or disconnected and reconnected the Wi-Fi/Ethernet).

6. Ran:

ipconfig /all

again to check the IPv6 address.

```
Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 1A-E3-47-C0-CB-2A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-E3-47-C0-CB-2A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Dell Wireless 1506 802.11b/g/n (2.4GHz)
   Physical Address. . . . . . . . . : 28-E3-47-C0-CB-2A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2402:a00:142:7323:cf4b:53ac:6dfd:1c78(Preferred)
   Temporary IPv6 Address. . . . . . : 2402:a00:142:7323:f876:f30a:3cac:aeb1(Preferred)
   Link-local IPv6 Address . . . . . : fe80::b1c8:55cf:38f0:d3b7%22(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.8(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, August 3, 2026 3:54:14 PM
   Lease Expires . . . . . . . . . . : Tuesday, August 4, 2026 4:41:15 PM
   Default Gateway . . . . . . . . . : fe80::1%22
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 254337863
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-78-CE-53-EC-F4-BB-22-32-15
   DNS Servers . . . . . . . . . . . : fe80::1%22
                                       8.8.8.8
                                       8.8.8.4
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Observation:

Observed Address Behavior: (if the address changed)

The temporary IPv6 address changed after restarting the computer or reconnecting to the network. This shows that the operating system generated a new temporary IPv6 address to improve user privacy.

Observed Address Behavior: (if the address did not change)

The temporary IPv6 address remained the same after restarting the computer or reconnecting to the network. This may happen because the temporary address is still within its valid lifetime and has not yet been replaced.

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Ans.:-

```
# Zomato-style Restaurant Search using IPv6

restaurants = [
    {"name": "Dominos Pizza", "ipv6": "fd00:1::101"},
    {"name": "McDonald's", "ipv6": "fd00:1::102"},
    {"name": "Pizza Hut", "ipv6": "fd00:1::103"},
    {"name": "KFC", "ipv6": "fd00:1::104"},
    {"name": "Subway", "ipv6": "fd00:1::105"}
]
search = input("Enter restaurant name: ").lower()
found = False
for restaurant in restaurants:
    if search in restaurant["name"].lower():
        print("Restaurant:", restaurant["name"])
        print("IPv6 Address:", restaurant["ipv6"])
        found = True
if not found:
    print("No matching restaurant found.")
```

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Ans.:-

IPv6 Subnetting Scenario and Configuration

AI-Generated IPv6 Subnetting Scenario

Organization: Food Delivery Company

Main IPv6 Network:

2001:db8:1000::/48

Subnet 1: Admin Department

Network: 2001:db8:1000:1::/64

Example Host Address: 2001:db8:1000:1::10

Subnet 2: Restaurant Department

Network: 2001:db8:1000:2::/64

Example Host Address: 2001:db8:1000:2::20

Subnet 3: Delivery Department

Network: 2001:db8:1000:3::/64

Example Host Address: 2001:db8:1000:3::30

Manual IPv6 Address Assignment

Selected Subnet:

2001:db8:1000:2::/64

Assigned IPv6 Address:

2001:db8:1000:2::20

Prefix Length:

/64

Default Gateway:

2001:db8:1000:2::1

Configuration Steps (Windows):-

1. Open Control Panel.

2. Go to Network and Sharing Center.

3. Click Change adapter settings.

4. Right-click the active network adapter and select Properties.

5. Select Internet Protocol Version 6 (TCP/IPv6) and click Properties.

6. Choose "Use the following IPv6 address".

7. Enter:

   IPv6 Address: 2001:db8:1000:2::20

   Subnet Prefix Length: 64

   Default Gateway: 2001:db8:1000:2::1

8. Click OK and close all windows.

9. Open Command Prompt and run:

   ipconfig

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:a00:142:7323:cf4b:53ac:6dfd:1c78
   Temporary IPv6 Address. . . . . . : 2402:a00:142:7323:f876:f30a:3cac:aeb1
   Link-local IPv6 Address . . . . . : fe80::b1c8:55cf:38f0:d3b7%22
   IPv4 Address. . . . . . . . . . . : 192.168.1.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%22
                                       192.168.1.1
```